One stroke. A smaller arm.

Bent box boom. Compact pivots. Inward bucket. Four identical 60 mm actuators.

Cat 320 proportions. One common actuator size.

R2 / 1:12
475 mm boom / 241.7 mm stick / 100 mm bucket



R2 strokes: 85 / 105 / 60 mm; four units in three sizes.

60 mm edition / 1:15
380 mm boom / 193.3 mm stick / 80 mm bucket



New strokes: 60 / 60 / 60 mm; four identical U60 units.

Same drawing scale and joint pose. Main reference dimensions are 20% smaller; pin hardware and linkage are adapted.

Published Cat 320 dimension	Reference	Model at 1:15
Reach boom pin centres	5700	380.000
R2.9 stick pin centres	2900	193.333
1.19 m3 HD bucket width	1200	80.000
Bucket tip radius	1570	104.667

Confirmed dimensions and adapted geometry

The four dimensions above come from Caterpillar technical specifications, pages 4, 5 and 11. Side photographs establish the shape and mechanism. Section outlines, detailed pin locations and the ground support are adapted to the V6 actuators; this is not factory CAD or a scaled factory load rating.

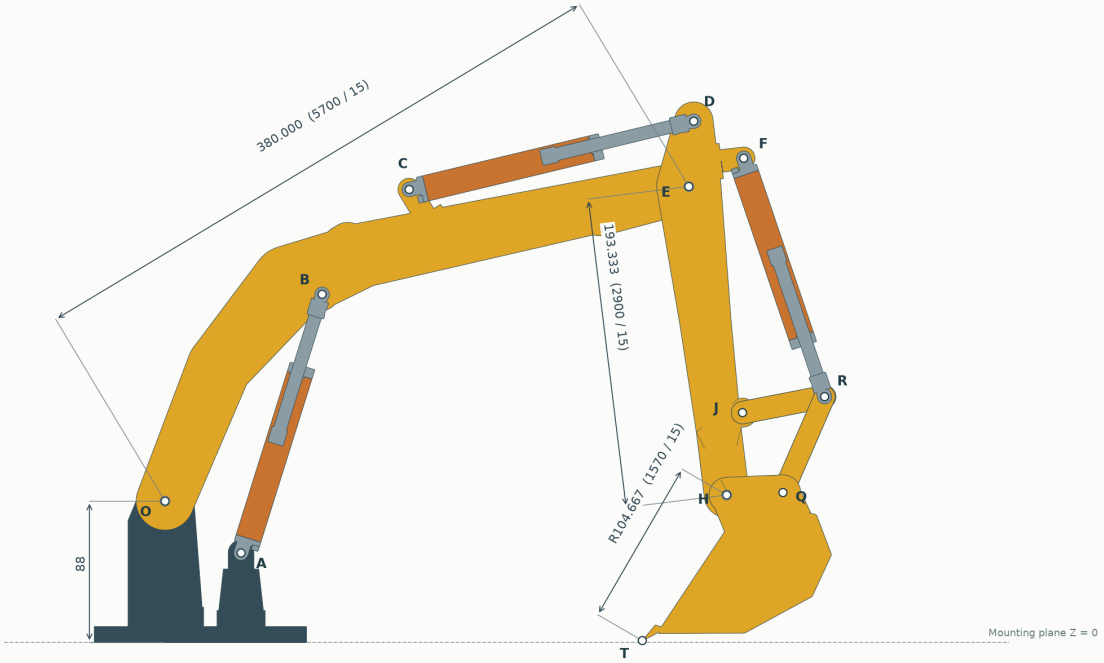
References: Finning Cat 320 product page; Caterpillar specifications and 360-degree side image. Full URLs and downloaded references are included under references/. Manufacturer images remain copyright Caterpillar.

Geometry & correct bucket action

Side silhouette is generated from the exported CAD meshes.

CAT 320 REFERENCE / V6 ARM / 1:15

DIMENSION SHEET | Actual CAD silhouette | Fixed base; no chassis | All dimensions in mm



Factory dimensions retained: boom 5700, stick 2900, bucket width 1200, tip radius 1570. Model bucket width: 80 mm.
Mounts, sections and cylinder pin placements are adapted geometry. J–R = 52; R–Q = 65; H–Q = 35. Not a cutting template.

Key interface	Model dimensions
Main pivots O, E, H	diameter 6 shaft / diameter 6.2 hole
Actuator and linkage pivots	diameter 3 shaft / diameter 3.2 hole
Four-bar J-R / R-Q / H-Q	52 / 65 / 35
Boom / stick box width	48 / 32
Base plate / root height	132 x 164 x 10 / 88

Extension must produce these actions

Boom rises. Stick crowds inward. Bucket cutting lip curls up toward the base.
Both boom cylinders share one angle and must remain equal in length.
The bucket mouth faces the machine in the reference pose. Its tip lies
approximately 1.3 mm above the mounting plane in that pose.

Actuators sized to the arm

One 75 mm screw and one 60 mm stroke actuator design serves all positions.

Function	Qty	Screw	Stroke	Closed-open
Boom, V6-U60	2	75	60	140-200
Stick, V6-U60	1	75	60	140-200
Bucket, V6-U60	1	75	60	140-200

Screw length is measured from gearbox front face to tip.
V6 closed pin spacing = screw length + 65; stroke = screw length - 15.
Each end retains 3 mm of mechanical reserve beyond nominal travel.
The original 55 mm V6-S screw and its PCB do not fit this 60 mm version.

Matching regenerated components

Build four copies of the same barrel, caps, rod and rod eye.
New end-stop boards, Gerbers, KiCad, BOM and placement files are included.
The common actuator CAD/STL checks pass; its board has zero design-rule errors and zero independent Gerber readback mismatch.

Rotation and electrical direction

M3 x 0.5 right-hand screw: 2 revolutions per mm.
Clockwise extends, viewed from the rod end toward the motor.
Nominal full strokes need 120 revolutions on every actuator.
Electrical A+ / B- extends; reverse polarity retracts. Nominal supply is 6 V.
There is no external position feedback from the original V6 circuitry.

Paired boom control

Use three reversible control functions; coordinate both boom cylinders together.
They represent one mechanical axis. Do not command unequal lengths.
Match both physical end stops before operation. The viewer sliders specify exact geometry; they are not a supplied hardware position controller.

The original electronics topology and magnetic assumptions are inherited.
The resized mechanical and PCB layouts were checked; powered operation, motor thrust capacity, temperature and real switch positions remain untested.

Assembly & verification

Use the matching sets and check the physical fits before loaded operation.

1. Anchor the fixed base using four M6 bolts through its diameter 6.6 holes.
2. Install boom O, stick E and bucket H with steel shafts and spacers.
3. Fit the paired rockers on J and outer links on R/Q on the illustrated branch.
4. Fit U60 cylinders at A-B (two), C-D and F-R. Keep boom strokes equal.
5. Fit collars with running endplay; add flexible external cable loops.
6. Jog unloaded and verify polarity, physical endpoints and paired reversal.

Measured checks on this revision

134 sampled CAD poses; 1021 distinct solid intersections checked.
No external solid interference at the sampled poses, including reserve corners.
461 transmission samples; maximum pin error below 0.000001 mm.
Minimum four-bar transmission angle: 33.16 degrees including reserve.
8 structural STL files: watertight and single-component; STEP imports pass.
The V6-U60 actuators, with their cable clamps, pass their own CAD and board checks.

Working envelope

The sampled cutting tip spans X 82.1 to 606.8 mm and
Z -195.4 to 524.7 mm relative to the mounting plane.
Below-plane poses represent excavation. A solid-floor rig must restrict these poses or provide a clear recess/bench edge. Sampled checks do not prove collision freedom at every continuous position.

Manufacturing and load limits

The boom STL is 414 mm long; it needs a large printer or an engineered split.
Closed hollow boxes require process-specific support and cavity planning.
A watertight model is not a validated load-bearing printed assembly.
Actual collar set screws, external wires and floor material are outside the collision model. Printing, alignment, rod deflection, PETG creep, gearbox thrust capacity and real end-stop accuracy need physical prototype tests.
No payload rating is supplied. See README.md for the shaft/spacer BOM.

Lift mounts integrated into the boom

R4: broader load paths at the two upper lift-cylinder connections.

R3 / narrow projecting supports



5 mm cheeks / 7 mm boss radius

R4 / broad saddles and cross-web



8 mm cheeks / 10 mm boss radius

Pins move 32.6 mm forward and 27.8 mm lower in boom coordinates. Both views use the same camera and pose.
60 mm stroke and 2–69° boom range retained. Geometry and motion checked; printed strength not load-tested.

Feature	Previous R3	Revised R4
Upper pin B, boom X / Z	117.372 / 87.808	150 / 60
Clevis cheek thickness	5	8
Boss radius / hole diameter	7 / 3.2	10 / 3.2
Upper lift pin arrangement	Two short pins	One diameter 3 x 126
Lower A, ground X / Z	55 / 53	47.277 / 55.771

Attachment and assembly changes

Broad rounded saddle feet join a solid web across both boom walls.
Both upper eyes share a steel cross-pin, retained by two outside collars.
Use the revised boom and fixed base together; keep the eye joints free.
The 60 mm actuators, 2-69 degree boom range and main arm sizes remain.

Print strength still needs physical validation

Print on the broad side as oriented in the STL. Use continuous perimeters and suitable local density around the bosses and saddle roots. Layer bonding, material, print settings and pin fit determine actual strength. No load rating or measured strength multiplier is claimed.